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Course: CS 418 - Fall 2024
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Project Proposal

- **Project name and Github (Paragraph 1) and Github Handle:**

- **Project's Name:** A Demographic Study of Illinois
- **The Description of Motivation:** To identify the factors that limit certain groups from attending college in Illinois, and explore solutions to bridge this gap.
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- **Team Members, UIC email handles, and github handles**
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- **Problem (Paragraph 2):**

This project's "big idea" aims to investigate the obstacles that keep some pupils from going to college while enabling others to do so. Finding patterns in socioeconomic level, racialization, gender, place of residence, and academic achievement is the aim. What can be done to address these factors and the reasons behind their influence on college attendance rates? Because education is essential to both societal advancement and economic mobility, this is an important topic. There will be numerous advantages to recognizing and resolving these differences, one of which will be a more equal educational system. I selected this issue because, in spite of efforts to solve it, inequality in college access persists. According to my hypothesis, financial constraints, resource scarcity, and other systemic issues make college admission more difficult for lower-income and minority populations.

- **Data (Paragraph 3):**

I intend to make use of information from openly accessible sources that include statistics on Illinois's high school graduation and college enrollment rates as well as demographic information. [College.ai](#) is another source to help people navigate choosing and paying for college. [The National Center for Education Statistics \(NCES\)](#) is another resource; it provides national datasets with Illinois-specific data. Additionally, I intend to investigate postsecondary enrollment trends using records from [the Illinois Board of Higher Education \(IBHE\)](#). These enormous databases, which may contain millions of rows, include a wide range of demographic and educational outcomes.

Spreadsheet of DataSet

- **Solution (Paragraph 4):**

In order to tackle this issue, I want to perform an exploratory data analysis (EDA) in order to pinpoint significant patterns and relationships between variables such as socioeconomic position, racial background, geographic region, and college attendance rates. Using machine learning methods like decision trees and logistic regression, I will create models that forecast which individuals are most likely to enroll in college given certain demographic characteristics. Because the system is interactive, users will be able to investigate the ways in which various factors impact college access. I intend to include an early version of the interactive interface and a prototype model that can display predictive results for various student demographics in the progress report.